

Generation Interconnection Request System Impact Study Report

For Queue Project Q21-1893

East Bismarck 100 kV
July 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1893. The Interconnection Customer proposes a 116.2 MW solar photovoltaic facility interconnecting at the East Bismarck 100 kV substation in the Illinois area. The request was received on September 30, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 508 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,083,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
E BISMARCK SUBSTATION – RIO BRAVO ROCKLIN 100.0 kV Circuit 1	TEXON – E BISMARCK 230.0 kV	100.9	189	26.7
RIO BRAVO ROCKLIN SUBSTATION – E BISMARCK 100.0 kV Circuit 2	TEXON – E BISMARCK 230.0 kV line	100.9	189	26.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E BISMARCK SUBSTATION – RIO BRAVO ROCKLIN 100.0 kV Circuit 1	Reconductor line	\$1,714,000	33% (\$568,000)
RIO BRAVO ROCKLIN – E BISMARCK 100.0 kV #2	Reconductor line	\$1,554,000	33% (\$515,000)
	Total allocated to project		\$1,083,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.